

Diagnostic Information Limits under Healthy Drift and Closed-Loop Memory

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Abstract—We study deterministic diagnostic calendars for a known stable linear plant with complete retained states and a scalar healthy input that drifts throughout holds and missing intervals. When the fixed Gaussian process-noise image contains the fault direction, the constant-force score loss across a missing span determines two regimes: a sharp deficit of order five halves when that loss is positive, and matching quadratic-order bounds when the score is preserved. Both results compare with an ideal healthy-subtracted complete record and use the complete input clock, calendar-uniform lower bounds, and observable attaining constructions. Positive-definite noise forces the first regime; a separate compact common-noise controller theorem permits joint leading-order design. Controller examples and an independent nonlinear mechanical certificate explain the information restrictions and finite-risk obligations of these laws.

I. INTRODUCTION

Known feedback transforms the fault, healthy input and process noise through the same plant. With complete state records, inverting that transformation preserves the diagnostic experiment. A missing state can prevent this inversion, so the diagnostic effect of a controller depends on which states are retained, rather than on output noise correlation alone.

We consider two task signs, a growing known fault, and a separate unrestricted-level healthy path under each hypothesis. The paths satisfy a rate bound throughout the complete input history, including every transition slot. The calendar and terminal horizon are fixed before the noise. Task changes therefore produce missing-data intervals with continued dynamics. The ideal healthy-subtracted complete-record information is cubic in the horizon; our question is the calendar's optimal deficit from that benchmark.

The missing-span loss of the constant-force score classifies this deficit. A positive loss gives an exact $H^{5/2}$ coefficient and a terminal-balanced calendar with linearly increasing holds. A zero loss gives $\Theta(H^2)$, without a sharp quadratic coefficient; its upper construction balances the complete cumulative task coefficient. Score preservation can occur with a fixed lower-rank noise image even when affine slope information is lost. Positive-definite process noise excludes it. The proof covers all legal calendars in the converse and repairs one scalar mass constraint in the genuine observable innovation range for attainment. A separate uniform theorem for compact controller families with common positive-definite noise permits optimization over both controller and calendar. A fixed bounded task-even bias preserves the positive coefficient and the quadratic order.

Controlled sensing and controlled Markovian observations already optimize observation actions for testing [1], [2]. Their

memoryless or stationary simple-hypothesis and sequential contracts differ from the present complete rate-limited paths, growing force, deterministic horizon and missing state intervals. Monetary switching-cost results also address distinct risks: Vaidhiyan and Sundaresan take the error limit before sending a sluggish-action parameter to zero [3]; Lambez and Cohen use known iid laws and sequential Bayes risk with switching penalty $s = O(c)$ as delay cost c tends to zero [4]. Full-history uncertainty sets and constrained active diagnosis also predate this work: Scott et al. separate complete output sets under bounded uncertainty [5], while Gaussian closest-convex-set detectors provide the fixed-calendar statistical interface used here [6]. The present contribution is the second-order calendar deficit within the stated experiment, rather than a new nearest-set detector or a first-order sensing law.

Feedback design that propagates both mean and covariance is established [7]; closed-loop set-valued observers preserve full-history diagnosis guarantees [8]. The feedback/active-diagnosis studies in [9], [10] are directly relevant precedents. Symmetric drift compensation and Allan-variance observing designs account for elapsed time and real dead time [11], [12], and minimax switchback designs optimize patterns under fixed horizons [13]. Their objectives and uncertainty contracts differ from ours, but these comparisons do not establish exhaustive historical priority. Full-text theorem comparisons remain unresolved for the two 2012 feedback papers and for Cheng–Steinberg's 1991 *Trend robust two-level factorial designs* and Coster–Cheng's 1988 *Minimum Cost Trend-Free Run Orders of Fractional Factorial Designs*. Their verified bibliographic identities do not establish noncoverage of the results proved here.

The information restrictions are substantive. The controller has the states needed for feedback, while the diagnostician reads only the declared retained states. Fast internal states, commands that reveal them, additional sensors, or diagnostic readings during transitions define different experiments. The blackout is a telemetry or admissibility restriction; feedback and healthy drift continue during it.

II. THE RETAINED-STATE EXPERIMENT

Consider the known physical plant and state feedback

$$\begin{aligned} x_j &= Ax_{j-1} + B_u u_j + B(\mathbf{1}_{\{h=1\}} a_j + g_j b_j^h) \\ &\quad + G \xi_j, \\ u_j &= -K x_{j-1}, \quad A_K = A - B_u K, \\ \xi_j &\stackrel{\text{iid}}{\sim} N(0, I_p). \end{aligned} \tag{1}$$

The dimension is fixed, A_K is strictly Schur, $G \in \mathbb{R}^{d \times p}$ has full column rank, and $0 \neq B \in \text{range } G$. Normality and $\|A_K\| < 1$ are unnecessary. Write $f = G^\top B$ and $F = \|f\|^2 > 0$. The known initial state is $x_{-n_0} = 0$, with fixed $n_0 \geq 1$. The prefix states at $j = -n_0 + 1, \dots, 0$ are all retained, with $g_j = 1$ and $a_j = 0$. For $j = 1, \dots, H$, $a_j = \rho_0 + \eta j$, where $\rho_0 \geq 0$ and $\eta > 0$. Throughout, $a_j^h = \mathbf{1}_{\{h=1\}} a_j$.

Each hypothesis $h \in \{0, 1\}$ selects a deterministic complete scalar healthy path b^h with unrestricted level and adjacent-input rate at most $r/2$, where $r > 0$. Nature observes the fixed deterministic calendar before choosing these paths and before the independent innovations are drawn. Their actual difference class is

$$\mathcal{Z}_H = \{z \in \mathbb{R}^{H+n_0} : |z_{j+1} - z_j| \leq r \text{ on every adjacent input pair}\}. \quad (2)$$

Holds retain the complete state in every slot at $g = \pm 1$. Changes exclude at least $k \geq 1$ consecutive state readings, with known $|g_j| \leq 1$ during the excluded slots. Known exact- k profiles in both directions can be repeatedly concatenated with holds at any planned start time; they need not be symmetric and may depend on that predetermined time. Longer transitions remain legal in the converse. Final holding and retained reads are allowed, and no terminal task is imposed. No state, noise or healthy level is reset. There is no hard actuator, state or tracking budget in this analytical class.

Let consecutive retained or initially known states be at t_{i-1}, t_i , with $m = t_i - t_{i-1}$. The true predictor is

$$\begin{aligned} x_{t_i} - A_K^m x_{t_{i-1}} &= \sum_{j=t_{i-1}+1}^{t_i} A_K^{t_i-j} \\ &\quad [B(a_j^h + g_j b_j^h) + G\xi_j]. \end{aligned} \quad (3)$$

Define its disjoint input map, covariance, and input projector by

$$\begin{aligned} H_m &= [A_K^{m-1}G, \dots, G], \\ W_m &= H_m H_m^\top, \\ P_m &= H_m^\top W_m^\dagger H_m. \end{aligned} \quad (4)$$

A gap with k excluded states has $k+1$ inputs, including its right retained input. Stack the genuine interval projectors to obtain P_π ; inputs after the last retained state have zero rows and columns in P_π . If U_m is an orthonormal basis for $\text{range } W_m$, then $T_m = (U_m^\top W_m U_m)^{-1/2} U_m^\top H_m$ supplies rank W_m nonredundant white rows. The stack T obeys $TT^\top = I$ and $T^\top T = P_\pi$.

With $\theta = (fa_j)_j$ and $H_g z = (fg_j z_j)_j$, define the profiled information and its deficit by

$$\begin{aligned} G_{K,H}(\pi) &= \frac{1}{2} \inf_{z \in \mathcal{Z}_H} \|P_\pi(\theta - H_g z)\|^2, \\ O_H &= \frac{F}{2} \sum_{j=1}^H a_j^2, \\ D_{K,H}(\pi) &= O_H - G_{K,H}(\pi), \\ D_{K,H}^* &= O_H - \sup_{\pi} G_{K,H}(\pi). \end{aligned} \quad (5)$$

The factor $1/2$ is the KL convention. Appendix A proves common support and this exact projection using the actual

predictor, without resetting the states or dropping cross-history covariance. O_H is the KL of the complete white record after ideal subtraction of the healthy input. It differs from the composite full-record information, which still profiles over z . Both the complete-record experiment and its oracle are neutral to known feedback. The supremum in (5) need not be attained at finite H .

III. TWO DEFICIT REGIMES

Theorem 1 (Fixed-plant deficit). *Under the experiment of Section II, put*

$$\begin{aligned} M_m &= \sum_{q=0}^{m-1} A_K^q B, \\ \beta_\ell &= (\ell+1)F \\ &\quad - M_{\ell+1}^\top W_{\ell+1}^\dagger M_{\ell+1} \geq 0. \end{aligned} \quad (6)$$

If $\beta_k > 0$, then

$$\begin{aligned} D_{K,H}^* &= \frac{\sqrt{2}}{5} \sqrt{F\beta_k r} \eta^{3/2} H^{5/2} \\ &\quad + o(H^{5/2}). \end{aligned} \quad (7)$$

For every fixed $\xi > 0$, the terminal-balanced calendar described below obeys

$$\begin{aligned} D_{K,H}(\pi_H^\xi) &\leq \frac{1}{10} \\ &\quad \left(\frac{2\beta_k \eta^2}{\sqrt{\xi}} + Fr\eta\sqrt{\xi} \right) H^{5/2} \\ &\quad + O_\xi(H^2), \end{aligned} \quad (8)$$

and attains the coefficient in (7) at $\xi_* = 2\beta_k \eta / (Fr)$. If $\beta_k = 0$, there exist fixed $0 < c < C < \infty$ and H_0 such that

$$cH^2 \leq D_{K,H}^* \leq CH^2, \quad H \geq H_0. \quad (9)$$

The first branch is leading-coefficient sharp; the second is order sharp and asserts no sharp quadratic constant. All constants are for the fixed plant, noise factor, controller and complete scalar path class.

The positive construction uses symmetric blocks with even n_ℓ : $n_\ell/2$ positive reads, k missing slots, n_ℓ negative reads, k missing slots, and $n_\ell/2$ positive reads. Their lengths increase as $n_\ell = \xi\ell + O(1)$; packing leaves one to four final positive reads. Appendix B defines the exact input starts, midpoints and half-slot gap offsets, proves the calendar-uniform converse, and constructs an observable direction whose single scalar mass is exactly zero. The zero branch instead uses a profile-aware calendar with bounded cumulative task coefficient, proved together with its all-calendar lower bound in Appendix C.

Proposition 1 (Preserved score and finite capacity). *Let $\mathcal{R}_m = \text{span}\{G, A_K G, \dots, A_K^{m-1} G\}$, $\mathcal{R}_\infty = \bigcup_m \mathcal{R}_m$, and $\nu_R = \min\{m : \mathcal{R}_m = \mathcal{R}_\infty\}$. Then $\beta_k = 0$ if and only if some $y \in \mathbb{R}^d$ satisfies*

$$G^\top (A_K^\top)^q y = f, \quad q = 0, \dots, k. \quad (10)$$

In this case

$$\begin{aligned} k < \nu_R &\leq \dim \mathcal{R}_\infty - p + 1 \\ &\leq d - p + 1, \quad k \leq d - p. \end{aligned} \quad (11)$$

This preserves the constant scalar force score; it does not require $P_{k+1} = I_{(k+1)p}$ or recovery of every latent input.

Corollary 1 (Positive-definite process noise). *If $Q = GG^\top \succ 0$, then $F = B^\top Q^{-1}B$ and $\beta_k > 0$ for every $k \geq 1$. Thus (7)–(8) give the fixed-order positive-definite-noise theorem.*

The score characterization, capacity bound and strict-positive specialization are proved in Appendix A.

Score preservation with lost slope information. For $k = 1$ under the same task protocol, take the nilpotent shift $A_K e_2 = e_1$, $A_K e_3 = e_2$, $A_K e_1 = 0$, with $G = [e_2, e_3]$ and $B = e_2 + e_3$. Then $f = (1, 1)^\top$, $F = 2$, and $H_2 = [e_1, e_2, e_2, e_3]$ has rank three in a four-dimensional input space. The constant ray $(1, 1, 1, 1)^\top$ is in its rowspan, so $\beta_1 = 0$, but $P_2 \neq I_4$. The centered affine ray $(-1/2, -1/2, 1/2, 1/2)^\top$ has residual norm squared $1/2$. Hence preserving the constant score can coexist with loss of slope and other input modes; the healthy-drift deficit still has quadratic order.

A. Fixed-calendar risk

The difference class $\mathcal{Z}_H$ is a finite-dimensional polyhedron, including its free initial level. Its linear image in the genuine whitened data coordinates is a closed polyhedron. The Euclidean closest difference therefore exists. A minimizing z^* realizes both healthy budgets through $b^0 = z^*/2$, $b^1 = -z^*/2$. With $\mathcal{B}_{r/2}$ the complete healthy class, the true mean sets are $\mathcal{M}_0 = TH_g \mathcal{B}_{r/2}$ and $\mathcal{M}_1 = T\theta + TH_g \mathcal{B}_{r/2}$. Choose $\mu_0^* = TH_g(z^*/2)$, $\mu_1^* = T\theta - TH_g(z^*/2)$, and put $d = \|\mu_1^* - \mu_0^*\| = \sqrt{2G_{K,H}(\pi)}$. For $d > 0$, nearest-pair convexity gives

$$\begin{aligned} (\mu_1^* - \mu_0^*)^\top (\mu_0 - \mu_0^*) &\leq 0, \\ (\mu_1^* - \mu_0^*)^\top (\mu_1 - \mu_1^*) &\geq 0 \end{aligned}$$

for every admissible mean under the respective hypothesis. The statistic $S = (\mu_1^* - \mu_0^*)^\top (\tilde{Y} - \mu_0^*)/d$ has unit variance and mean at most zero under the null, at least d under the alternative. Thresholding at $z_{1-\alpha}$ has worst false alarm at most α and worst power at least $\Phi(d - z_{1-\alpha})$. Here $\tilde{Y}$ is the stack of whitened true conditional innovations. The realizing simple pair and the Neyman–Pearson bound show this power is optimal among tests with the same uniform false-alarm constraint. Thus, for $0 < \alpha, \beta < 1$ and $\alpha + \beta < 1$, power at least $1 - \beta$ is possible exactly when

$$G_{K,H}(\pi) \geq \frac{1}{2}(z_{1-\alpha} + z_{1-\beta})^2.$$

If $d = 0$ the closed sets overlap, so no test can have uniform power exceeding its uniform false-alarm allowance. This is the classical Gaussian closest-set interface [6], not an additional calendar theorem.

IV. CONTROL CONSEQUENCES

When comparing gains, write $\beta_K(\ell)$ for the penalty in (6) computed with that gain. The original force and noise are common throughout each controller comparison.

A. Equal poles do not fix diagnostic loss

Take the same physical plant and process noise

$$\begin{aligned} A &= \text{diag}(4/5, 3/5), \quad B_u = I_2, \\ B &= (1, 2)^\top, \quad Q = \begin{pmatrix} 1 & 1/3 \\ 1/3 & 5/9 \end{pmatrix}. \end{aligned}$$

Here $F = 29/4$. Gains $K_d = \text{diag}(3/10, 7/20)$ and $K_n = \begin{pmatrix} 3/10 & -2 \\ 0 & 7/20 \end{pmatrix}$ give closed matrices $\text{diag}(1/2, 1/4)$ and $\begin{pmatrix} 1/2 & 2 \\ 0 & 1/4 \end{pmatrix}$. They share the poles and full oracle, but at $k = 1$,

$$\begin{aligned} \beta_{K_d}(1) &= 1343/344, \\ \beta_{K_n}(1) &= 18007/10456. \end{aligned}$$

The ratio of sharp deficit coefficients is approximately 0.664. This is an ideal known-gain comparison with common B, Q and no actuator budget; it does not identify a hardware-optimal controller.

B. Settling time changes the controller comparison

For a scalar common-input plant with $Q = q > 0$, $B = 1$ and closed parameter $0 \leq c < 1$, write

$$\begin{aligned} A_m(c) &= \sum_{j=0}^{m-1} c^j, \\ B_m(c) &= \sum_{j=0}^{m-1} c^{2j}, \\ \kappa_c(k) &= k + 1 - A_{k+1}(c)^2 / B_{k+1}(c). \end{aligned}$$

Thus $F = 1/q$, $\beta = \kappa/q$ and the coefficient is $\sqrt{2}\sqrt{\kappa r \eta}^{3/2}/(5q)$. At fixed $k \geq 1$, κ strictly decreases in $c \in [0, 1)$. To impose a common homogeneous qualification, let $|e_0| \leq E$, $e_{j+1} = ce_j$, $\gamma = \epsilon/E \in (0, 1)$ and a fixed integer move duration $d_{\text{move}} \geq 0$. Require the smallest positive s with $c^s \leq \gamma$, and set $k(c) = d_{\text{move}} + s(c)$. This requirement concerns the homogeneous component; forcing and innovations continue during the excluded slots. It does not assert full noisy return to a tube.

Proposition 2 (Plateau candidates). *On a compact $0 < c_{\text{lo}} \leq c \leq c_{\text{hi}} < 1$, the minimum of $\kappa_c(k(c))$ is achieved in the finite set*

$$\begin{aligned} &\{c_{\text{hi}}\} \cup \\ &\{\gamma^{1/s} : c_{\text{lo}} \leq \gamma^{1/s} \leq c_{\text{hi}}, \\ &\quad 1 \leq s \leq s(c_{\text{hi}})\}. \end{aligned}$$

If $L = -\log \gamma$ and $\delta = -\log c$, then

$$\begin{aligned} \delta \kappa_c(k(c)) &\longrightarrow L - 2 \tanh(L/2) > 0 \\ &(c \uparrow 1). \end{aligned}$$

The proof is in Appendix D.

For $\gamma = 81/100$ and $d_{\text{move}} = 0$, the finite controller family $\{4/5, 9/10, 19/20\}$ has $k = 1, 2, 5$ and exact gap penalties $1/41, 2/91, 5064645/111045881$. The middle gain is best in that family. On the full interval $[4/5, 19/20]$, the unique best is instead $c = 81/100$, with $\kappa = 361/16561$. The latter ranking uses rational isolation of the irrational plateau endpoints. It optimizes the declared single-gap or coefficient function.

Pointwise calendar sharpness alone does not exchange the limit with optimization over an entire continuous controller family. For a finite fixed-gain family, the maximum of finitely many little-oh remainders is still little-oh.

C. Joint leading-order design under common positive-definite noise

Theorem 2 (Compact common-noise controller family). Write $D_H^{*,K} := D_{K,H}^*$ and $D_H^{\text{joint}} := O_H - \sup_{K,\pi} G_{K,H}(\pi) = \inf_K D_H^{*,K}$. Here Π_K is the legal calendar class for gain K , and the joint supremum uses $K \in \mathcal{K}$, $\pi \in \Pi_K$. Let the fixed plant, $B \neq 0$, $Q \succ 0$, healthy rate, fault template, prefix and diagnostic information be common. Let $\mathcal{K}$ be a nonempty compact gain set for which each A_K is strictly Schur, and let $k(K) \in \{1, \dots, k_{\max}\}$. For each gain assume repeatedly concatenable known exact- $k(K)$ transitions in both directions, bounded by $|g| \leq 1$. Set $C(K) = \sqrt{2} \sqrt{F\beta_K(k(K))r\eta^{3/2}}/5$. Then

$$\sup_{K \in \mathcal{K}} |H^{-5/2} D_H^{*,K} - C(K)| \rightarrow 0,$$

and $D_H^{\text{joint}}/H^{5/2}$ tends to $\inf_K C(K)$.

The uniform proof is in Appendix D.

No continuity of $k(K)$ or of its known profiles is needed for this limit; neither ensures a finite-horizon optimizer or a minimum of $C(K)$. In the declared scalar settling example, the finitely many plateau endpoints are lower semicontinuous because the shorter k is included at each equality. Its unique coefficient minimum is $c_* = 81/100$. Theorem 2 therefore makes that fixed controller and its terminal-balanced calendar jointly optimal at leading order under the same homogeneous qualification. Any sequence of controllers within $o(H^{5/2})$ of the optimal deficit converges to c_* : on the compact set outside each neighborhood, lower semicontinuity and uniqueness give a positive coefficient gap, while the information remainder is uniform. This adds a joint asymptotic conclusion, not a computed finite- H risk optimum or complete forced-return guarantee.

V. BOUNDED TASK-EVEN UNCERTAINTY

Corollary 2. Add Be_j^h with $|e_j^h| \leq E/2$ to the input, where E is fixed. Assume the complete difference class is the explicit Minkowski envelope $gZ_H + \mathcal{E}_H$, with $0 \in \mathcal{E}_H$ and $|e_j| \leq E$, and leave the covariance, plant and calendar class unchanged. With $N = H + n_0$ and $A_H = (\sum a_j^2)^{1/2}$, every calendar satisfies

$$0 \leq G_{\text{odd}}(\pi) - G_{\text{mix}}(\pi) \leq FE\sqrt{N}A_H = O(H^2).$$

The same bound holds for $D_{\text{mix}}^* - D_{\text{odd}}^*$ under the same oracle. Thus the positive leading coefficient is unchanged. Under the fixed common-support quadratic branch of Theorem 1, the same actual envelope and unchanged projection give $D_{\text{odd}}^* \leq D_{\text{mix}}^* \leq D_{\text{odd}}^* + O(H^2)$, so the order remains quadratic without preserving a quadratic coefficient.

The contraction proof is in Appendix E.

This is an order-two perturbation bound, not an exact constant translation. Multiplicative uncertainty that changes

the plant, covariance or support is a different model. If an unrestricted even difference class contains a , choosing $e = a$ and $z = 0$ makes the hypotheses indistinguishable.

VI. FINITE VERIFICATION AND A NONLINEAR RISK ILLUSTRATION

Exact rational checks compare the initialized state covariance against the retained-input projector in scalar and matrix examples. The matrix report contains 150 retained-subset checks and 68 full calendars, including rotating, nonnormal and Jordan models. The proof is universal; these finite checks support implementation. The symmetric transition profiles in the original formal reports have zero total repair, so they do not numerically exercise nonzero α . A separately reviewed asymmetric matrix example has $N = -22863/1307$, $D = 628321/10456$ and $\alpha = -182904/628321$. Its repaired healthy mass is exactly zero. The separately reproduced six-case asymmetric supplement includes two low-rank examples with repair energies 100/77 and 507/178. These supplement the original zero-repair checks without changing their historical counts.

The separate planar six-joint mechanical certificate uses a fixed joint-4 fault $f(t) = 0.0003(t-1)_+$, four healthy prefix slots, fast step 0.001 s, and diagnostic slot 0.25 s. Move and settle exclude twelve slots while all process and encoder innovations continue. Point readout is the last existing pre-step fast sample in the slot; average readout uses its 250 samples. The readout definitions and rounding were fixed before evaluation. This comparator has parameter-dependent physical bounds and deterministic nonlinear approximation errors; it is not claimed to be a direct exact instance of the fixed common-covariance matrix theorem.

The certificate targets are $\text{PFA} < 10^{-3}$ and protected power at least 0.9; $H = 120$ means 31 seconds including the four prefix slots. The direction is the frozen nodal projection of the declared terminal-balanced template, converted to physical task-signed weights and normalized by a conservative rational norm bound. Exact weights and their KKT/source identities are supplied with the case certificates; they are chosen before the data. For every allowed deterministic parameter/path ϑ under hypothesis h , the fixed physical statistic Z has a coupled unconditional Gaussian comparator $Z_G^{h,\vartheta} \sim N(\mu_{h,\vartheta}, V_{h,\vartheta})$. The bounds $\mu_{0,\vartheta} \leq m_0$, $\mu_{1,\vartheta} \geq m_1$, $V_{h,\vartheta} \leq V^+$, $|Z - Z_G^{h,\vartheta}| \leq e_h$ on the good event, and bad-event probability at most δ_h hold uniformly over the frozen instance class. The synthetic fast noises are independent Gaussian modeling choices: torque variance 10^{-4} and encoder variance $(2\pi/2^{17})^2/12$; they are not measured hardware distributions. The event can depend on the same innovations: the proof uses event inclusion and a union bound, not a conditionally Gaussian assumption. Set

$$t = m_0 + e_0 + \frac{25}{8}\sqrt{V^+},$$

$$g_{\text{prot}} = m_1 - e_1 - t.$$

“Protected gap” denotes g_{prot} . With $\bar{\Phi}$ a conservative upper bound on the standard normal tail, event inclusion and the unconditional comparator law give $\text{PFA} \leq \bar{\Phi}(25/8) + \delta_0$: on the null good event, $Z > t$ implies $Z_G^{0,\vartheta} > t - e_0$. On the

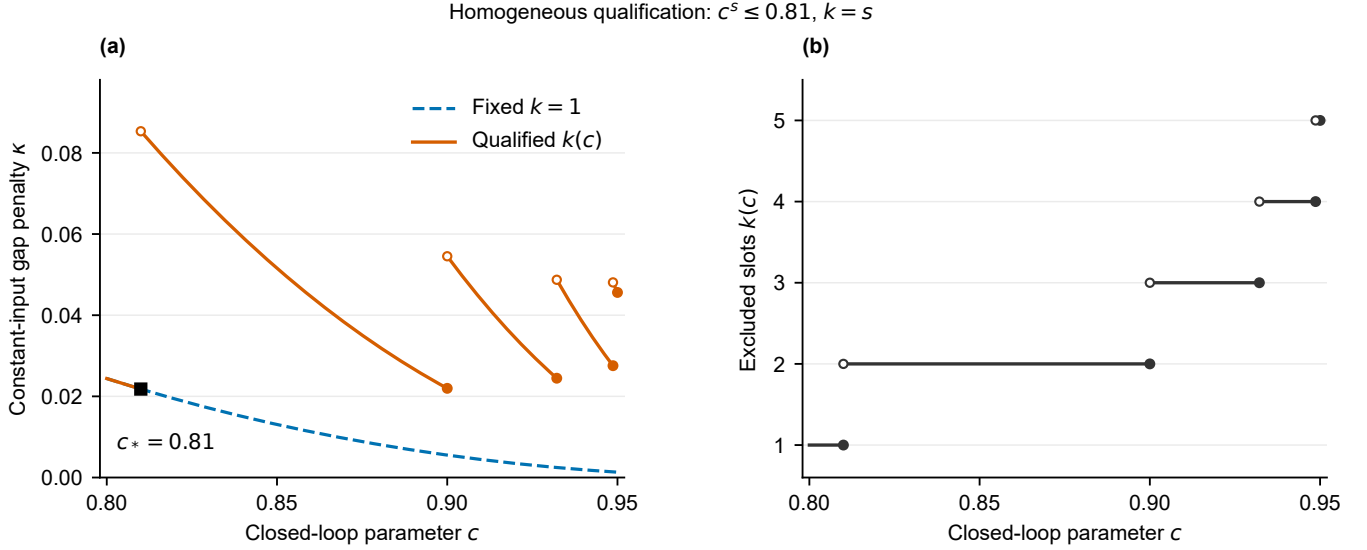


Fig. 1. Memory benefit after paying the same homogeneous settling requirement. The common scalar plant has innovation variance q and a known force input; κ is the dimensionless constant-input missing-span penalty ($\beta = \kappa/q$). For $c \in [.8, .95]$, the qualification uses $E = 1$, $\epsilon = .81$, no extra move slots, and the smallest positive integer s with $c^s \leq .81$; $k = s$. (a) The fixed- $k = 1$ penalty decreases smoothly, whereas the qualified penalty has downward segments and upward right-side jumps. The black square marks the exact continuum minimum at $c = 81/100$, $\kappa = 361/16561$. (b) The same qualification determines the excluded-slot count. Filled right endpoints belong to the shorter k ; hollow left endpoints show unattained right limits. Irrational endpoint positions are rounded only for display. The exact three-point family $\{.8, .9, .95\}$ instead has its best member at $.9$; the full-interval ranking comes from rational candidate certificates, not curve resolution. These are analytical curves with no random observations or confidence intervals. The qualification constrains only the homogeneous component and does not certify full noisy/forced return, nonlinear tracking or actuator feasibility.

alternative good event, $Z \leq t$ implies $Z_G^{1,\vartheta} \leq t + e_1$. When $g_{\text{prot}} > 0$, protected power is at least $1 - \Phi(g_{\text{prot}}/\sqrt{V^+}) - \delta_1$, using a conservative upper normal-tail bound Φ . When the gap is nonpositive this frozen sufficient certificate returns zero, without substituting the upper variance into a negative-gap power calculation. The displayed table values are rounded; exact rational/interval certificate endpoints, not these display decimals, are used for risk calculation. For the average case each deterministic allowance is approximately 0.000874611 and each dynamic mean loss is 0.000653168 ; for the point case they are 0.0177802 and 0.00695240 . The mean endpoints already include the dynamic loss on both sides, and each bad-event budget is approximately $2.06634 \cdot 10^{-11}$. The numerical certificate records bind the exact direction, both mean/error/event budgets, and the complete-history covariance witnesses to this fixed case. This presentation does not borrow the exact common-covariance nearest-set formula to certify the nonlinear plant.

For the finished terminal-balanced $H = 120$ case, complete-history variance certificates and two-sided mean/error/event budgets give

Readout	Upper variance	Protected gap	Power lower bound
250-sample average	$6.7073942 \cdot 10^{-7}$	0.0111325	0.999999999979
Last existing point	0.0181313	-0.453312	0

Both false-alarm upper bounds are approximately

0.000967122592 . The point row fails the sufficient power requirement; its lower bound zero is not a statement that actual power is zero. The inherited equilibrium enclosure is restricted to the frozen ramp through $H \leq 600$ and actual $f \leq 0.045$, not for every signal in the wider historical force box. Figure 2 extends this finite comparison over the complete declared terminal-balanced subgrid $H \in \{40, 60, 80, 100, 120, 160, 200\}$. Every nondegenerate row is retained, including failed sufficient power certificates. The two $H = 40$ directions are degenerate and have no risk bound. The first certified average-readout grid point for this prescribed family is $H = 100$, or 26 seconds including the prefix. This is neither a sequential stopping time nor an optimum over calendars. The declared 400-row family is not completely certified; this closed subgrid does not complete that calculation.

VII. LIMITS AND CONCLUSIONS

The two calendar laws require a known fixed stable plant and controller, complete retained states, a fixed full-column Gaussian noise factor, and a nonzero scalar fault/healthy force in its image. They do not cover noisy partial readings, unknown matrices or onset, arbitrary vector nuisance, saturation or adaptive calendars. Finite healthy amplitude boxes have a different long-horizon regime. The uniform controller theorem additionally requires common positive-definite process noise and compactness; near-singular or rank-changing families require separate analysis. Homogeneous settling is not a guarantee of full forced or noisy return. The nonlinear

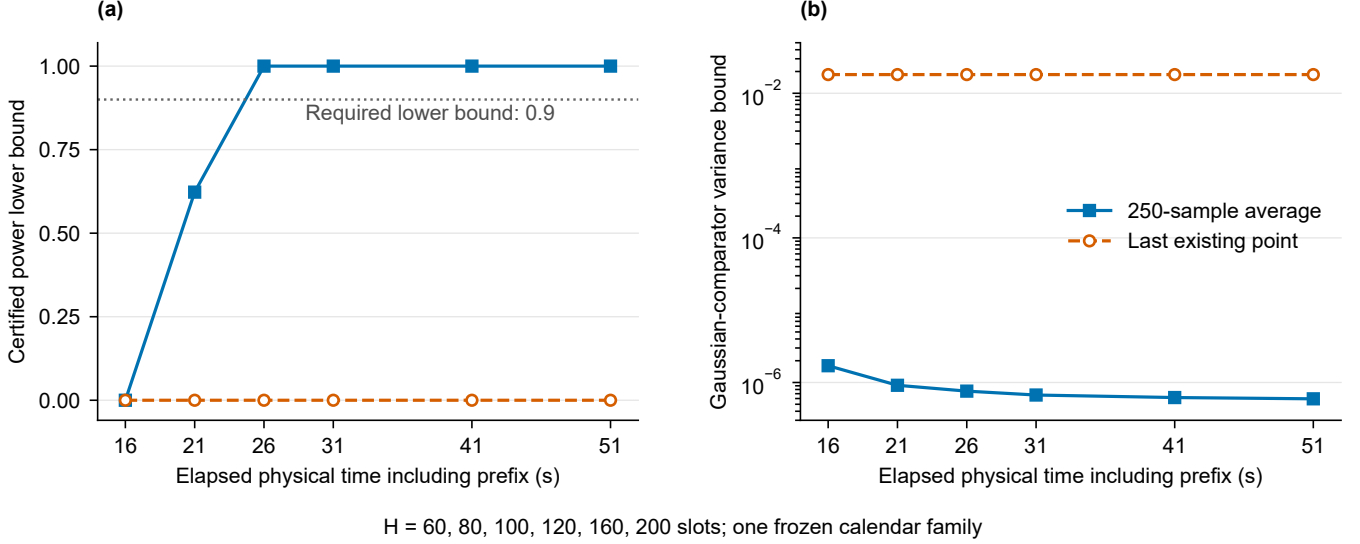


Fig. 2. Finite certificates for paired readouts of a prescribed terminal-balanced calendar family. The frozen nonlinear example uses four healthy prefix slots, 0.25-second slots and the fixed template $f(t) = 0.0003(t - 1)_+$. Each horizon has its own declared calendar, shared by its two readouts. (a) Uniform protected-power lower bounds of the prescribed statistics; all plotted false-alarm upper bounds are below 10^{-3} . The first certified average-readout grid point in this family is $H = 100$. The point bounds are zero because the sufficient certificate gives no positive guarantee; actual power is not asserted to be zero. (b) Complete-history Gaussian-comparator variance upper bounds for these unstandardized statistics. Directions use the prescribed rational norm scaling, while the statistics are not divided by their standard deviations. Their weights depend on the horizon and readout; the variance ratio alone does not quantify universal readout improvement. The true nonlinear statistics are controlled separately through remainder and event allowances. The two $H = 40$ directions have no risk certificate and are omitted from the numeric axes. Lines only connect certified grid values; these are bounds, not measured frequencies or confidence intervals. This finite comparison does not instantiate the linear asymptotic theorem or complete the remaining full-grid calculation.

illustration certifies only its stated fixed instance and provides no hardware optimum.

Within this experiment, the constant-force score loss separates a sharp $H^{5/2}$ deficit from an order-sharp quadratic deficit. The observable calendar constructions expose how controlled memory, complete healthy drift and missing time interact. The controller results turn that information law into a leading-order design rule under common physical inputs and timing.

APPENDIX A

COMMON SUPPORT, INTERVAL GEOMETRY AND PATH DUALITY

In the proofs, $P = P_\pi$ denotes the actual input projector, $\theta = (fa_j)_j$, and $\gamma = (g_j f)_j$. All sums include the fixed prefix unless a post-input range is specified.

Projection, score characterization and capacity. Let $\mathcal{F}_{A_K}$ be the initialized causal map from process inputs to states, and R retain the declared states. Their noise map is $L_R = R\mathcal{F}_{A_K} \text{diag}(G)$. Every mean under either hypothesis is $L_R \vartheta$ with $\vartheta = (f(a_j^h + g_j b_j^h))_j$. All laws consequently have the same support range L_R . For the reduced SVD $L_R = U_s \Sigma_s V_s^\top$, the data on this support are equivalent to $\Sigma_s^{-1} U_s^\top Y$, with standard Gaussian noise and mean $V_s^\top \vartheta$. Thus their KL is precisely $\|V_s^\top (\vartheta_1 - \vartheta_0)\|^2/2$.

The predictor transform in (3) is invertible and block triangular, with identity diagonal blocks. It uses the actual previous retained state, including the known initial state for the first interval. It changes L_R into disjoint interval maps H_m and preserves its rowspace. Therefore P_m is an orthoproject and

(5) is the exact profiled KL. This uses rowspace invariance, not an invalid congruence rule for pseudoinverses. If U_m is an orthonormal basis matrix for range W_m , the nonredundant white rows are $(U_m^\top W_m U_m)^{-1/2} U_m^\top H_m$, with identity covariance of dimension rank W_m . The full record instead recovers $G^\dagger(x_j - A_K x_{j-1})$, giving the stated control-neutral oracle.

We also need a bound valid for all span lengths. Cayley–Hamilton makes $\mathcal{R}_d$ invariant and $\mathcal{R}_m = \mathcal{R}_d$ for $m \geq d$. For such m , $W_m \succeq W_d$ with the same kernel, so inverse monotonicity on $\mathcal{R}_d$ gives $W_m^\dagger \preceq W_d^\dagger$. Hence

$$\begin{aligned} \Gamma_K &= \max_{1 \leq m \leq d} \|W_m^\dagger\| < \infty, \\ \sup_{m \geq 1} \|W_m^\dagger\| &\leq \Gamma_K, \\ S_B &= \sum_{q \geq 0} \|A_K^q B\| < \infty, \\ C_K &= \Gamma_K S_B^2. \end{aligned} \tag{12}$$

The last summability follows from strict stability, without normality or $\|A_K\| < 1$. The inverse bound does not use $W_1^\dagger$ on subspaces reached only later.

With $d_m = \mathbf{1}_m \otimes f$, $H_m d_m = M_m$ and

$$\beta_{m-1} = \|(I - P_m) d_m\|^2.$$

Zero loss is exactly $d_m \in \text{row } H_m$, which is (10). The sequence β_ℓ is nondecreasing; on a common initialized horizon of $m + 1$ constant-shift inputs, retaining times $1, m + 1$ has norm loss β_{m-1} , while deleting the time-1 state has

loss β_m ; rowspace inclusion makes loss nondecreasing. Also $0 \leq M_m^\top W_m^\dagger M_m \leq C_K$, so

$$\beta_\ell \geq (\ell + 1)F - C_K, \quad \beta_\ell/\ell \longrightarrow F. \quad (13)$$

Before reachability stabilizes, its dimension grows by at least one: $\mathcal{R}_{m+1} = \mathcal{R}_m$ already implies invariance. Since $\dim \mathcal{R}_1 = p$, this proves the upper bounds on ν_R . If (10) holds, set $\zeta = (A_K^\top - I)y$. Then $\zeta \perp \mathcal{R}_k$. If $k \geq \nu_R$, this annihilates the invariant reached space, and $G^\top (A_K^\top)^q y$ has zero successive differences for every $q \geq 0$. It remains the nonzero vector f , whereas strict stability makes it tend to zero. This contradiction proves (11).

For the positive-definite specialization, G is square invertible. If $m \geq 2$ and $H_m^\top y = d_m$, its last two blocks give $G^\top y = f$ and $G^\top A_K^\top y = f$. Thus $y = Q^{-1}B \neq 0$ and $A_K^\top y = y$, contradicting strict stability. This proves Corollary 1 for every nonempty internal gap. $\square$

Affine spans without reflection symmetry. For a span of length m , write $q_j = j - (m + 1)/2$, $d_q = (q_j f)_j$, $N_m = \sum_{j=1}^m q_j A_K^{m-j} B$. At center amplitude $A \geq 0$ its exact oracle norm loss is

$$\begin{aligned} & \|(I - P_m)(Ad_m + \eta d_q)\|^2 \\ &= \beta_{m-1} A^2 + 2\eta A L_{1,m} + \eta^2 L_{2,m}, \\ & L_{1,m} = -M_m^\top W_m^\dagger N_m, \\ & L_{2,m} = \|(I - P_m)d_q\|^2 \geq 0. \end{aligned} \quad (14)$$

By (12), $|L_{1,m}| \leq mC_K/2$. Thus the negative cross terms over any calendar cost at most

$$E_H = C_K \eta (\rho_0 + \eta H) H = O(H^2). \quad (15)$$

No reflection symmetry is needed.

The unprojected constant-slope inner product is zero because $\sum_j q_j = 0$. This justifies the stated $L_{1,m}$ even for nonnormal matrix dynamics; it does not set the projected cross term to zero.

Complete-path support and an observable dual. For scalar weights on the complete input clock define $q_t(\lambda) = \sum_{j=-n_0+1}^t \lambda_j$. Writing each path as its free first level plus adjacent increments and summing by parts gives

$$h_{\mathcal{Z}_H}(\lambda) = \begin{cases} r \sum_{t=-n_0+1}^{H-1} |q_t(\lambda)|, & \sum_j \lambda_j = 0, \\ +\infty, & \sum_j \lambda_j \neq 0. \end{cases} \quad (16)$$

Indeed the increment at $t + 1$ has coefficient $-q_t$, so the independently allowed increments attain the finite sum by choosing $\Delta z_{t+1} = -r \operatorname{sign} q_t$. A nonzero total coefficient makes the free first level unbounded. The same formula on a local block uses its actual consecutive input slots, including missing ones.

For $Pv = v$ let $\lambda_j = \gamma_j^\top v_j$. On each actual span the coefficient $c_m = W_m^\dagger H_m v_m$ acting on the predictor innovation has latent coefficient $H_m^\top c_m = P_m v_m = v_m$, mean $v_m^\top \theta_m$ and variance $\|v_m\|^2$. Hence this range vector is a statistic

of retained states alone, also for singular W_m . Applying $\|u\|^2/2 \geq v^\top u - \|v\|^2/2$ before the healthy-path infimum yields

$$\begin{aligned} G_{K,H}(\pi) &\geq \langle v, \theta \rangle - \frac{1}{2} \|v\|^2 \\ &\quad - h_{\mathcal{Z}_H}(\lambda), \\ D_{K,H}(\pi) &\leq \frac{1}{2} \|\theta - v\|^2 + h_{\mathcal{Z}_H}(\lambda) \\ &= \frac{1}{2} \|(I - P)\theta\|^2 + \frac{1}{2} \|P\theta - v\|^2 \\ &\quad + h_{\mathcal{Z}_H}(\lambda). \end{aligned} \quad (17)$$

No data from an unobserved input or fast state enter this dual.

APPENDIX B

POSITIVE-SCORE-LOSS CONVERSE AND ATTAINMENT

Positive branch of Theorem 1. Assume $\beta_k > 0$. Monotonicity and (13) imply $b_K = \inf_{\ell \geq k} \beta_\ell/\ell > 0$: with $L_0 = \max\{k, \lceil 2C_K/F \rceil\}$, the tail and finite initial part give $b_K \geq \min\{F/2, \beta_k/L_0\} > 0$.

Full-history calendar-uniform converse. Replacing terminal moves without later readings by holding preserves the old data and adds readings. Data processing for each full healthy pair and then an infimum show that the supremum can be restricted to calendars with a retained state at H . Prefix states are all read, so every non-singleton interval is post-onset and internal.

Fix $\varepsilon \in (0, 1)$ and divide $[\varepsilon H, H]$ into complete macros of length $L = \lfloor H^{3/4} \rfloor$. On each clipped observed run of length n , index its nodes by $i = 0, \dots, n-1$ and use the scalar signed tent $z_i = g_i r \min(i, n-1-i)$. Set it to zero on missing inputs, right-first retained inputs, run endpoints, prefix, macro joins and unused tail. This is one full rate- r path, realized by $z/2, -z/2$ under the two hypotheses. Writing $h = gz$, every non-singleton input span has residual $a - h = a$. If T is the sum of its actual gap norm losses, orthogonal decomposition gives

$$2D_{K,H}(\pi) \geq F \sum_j (2a_j h_j - h_j^2) + T. \quad (18)$$

No stationary endpoint or prefix replacement is needed.

In a macro let its minimum amplitude be A , its number of missing slots d , its clipped run count M and lengths n_i , with $\sum n_i = L - d$. Put $C = r(2A - rL/2)/4 > 0$ for large H . For a run, $\sum_{i=0}^{n-1} \min(i, n-1-i) \geq n^2/4 - n/2$ and each tent height is at most $rL/2$. Thus its saving is at least $Cn^2 - 2Cn$, and the macro saving is at least $C \sum n_i^2 - 2CL$. There are $M - 1$ complete actual gaps wholly within the macro, each excluding at least k slots and having center amplitude at least A . Equation (14) and monotonicity give

$$T \geq \sum_{\text{macros}} \beta_k A^2 (M - 1) - E_H.$$

For each gap its missing amplitudes are at most twice its input-center amplitude; hence the same T also satisfies

$$T \geq (b_K/4) \sum_{\text{macros}} A^2 d - E_H.$$

Crossing gaps contribute no complete-gap count but each missing node is assigned once in this second budget.

Fix $\delta \in (0, 1)$ and split the same global T before applying its bounds. Set $W = (1 - \delta)\beta_k$. For $M > 0$, Cauchy and AM-GM lower-bound a macro contribution by

$$\begin{aligned} & FC(L - d)^2/M + WA^2M + \delta b_K A^2 d/4 \\ & - WA^2 - 2FCL \\ & \geq 2A\sqrt{FCW}(L - d) + \delta b_K A^2 d/4 \\ & - WA^2 - 2FCL. \end{aligned}$$

The condition $A \geq 32FWr/(\delta^2 b_K^2)$ absorbs its missing-slot term, uniformly on late macros. For $M = 0, d = L$, the dead budget gives the same bound $2A\sqrt{FCW}L - WA^2 - 2FCL$ without division by zero. The cross error is paid once after the convex split.

As $A = \Theta(H)$ and $C = (rA/2)(1 + O(H^{-1/4}))$, the leading sum is $\sqrt{2FrW} \sum A^{3/2}L$. Coefficient, rounding and incomplete Riemann-block errors are $O(H^{9/4})$, $\sum A^2 = O(H^{9/4})$, $\sum CL = O(H^2)$ and $E_H = O(H^2)$. All constants are calendar-independent. The Riemann sum equals $(2/5)\eta^{3/2}(1 - \varepsilon^{5/2})H^{5/2} + o(H^{5/2})$. Taking the optimal deficit before the limit and then $\varepsilon, \delta \downarrow 0$ proves

$$\liminf D_{K,H}^*/H^{5/2} \geq \frac{\sqrt{2}}{5} \sqrt{F\beta_k r} \eta^{3/2}. \quad (19)$$

Packing and genuine singleton joins. A symmetric block with even $n = 2h$ has h reads at $+$, k missing slots, n reads at $-$, k missing slots and h reads at $+$. For block ℓ define its length, preceding post-input count, input midpoint and center amplitude by

$$\begin{aligned} L_\ell &= 2(n_\ell + k), \quad s_\ell = \sum_{i < \ell} L_i, \\ c_\ell &= (L_\ell + 1)/2 = n_\ell + k + \frac{1}{2}, \\ A_\ell &= \rho_0 + \eta(s_\ell + c_\ell). \end{aligned} \quad (20)$$

Its inputs are $s_\ell + 1, \dots, s_\ell + L_\ell$. For local scalar weights write $q_t(\lambda) = \sum_{j=1}^t \lambda_j$, $1 \leq t < L_\ell$. With budget $H - 1$, fit the maximal number of blocks with $\bar{n}_\ell = 2\lceil \xi\ell/2 \rceil$; evenly distribute remaining multiples of four slots, adding two to n_ℓ per unit and using a fixed left-to-right tie rule. Observe the last one to four states at $+$. Every block padding is $O(1)$ and

$$\begin{aligned} M &= \sqrt{H/\xi} + O(1), \quad n_\ell = \xi\ell + O(1), \\ A_\ell &= \rho_0 + \eta(s_\ell + c_\ell) \\ &= \eta\xi\ell^2 + O(\ell + 1). \end{aligned}$$

At an artificial join both adjacent states are actually read, so the next conditional input interval is a singleton. The global input projector therefore splits into its genuine block spans. It does not reset any state or make output covariances independent.

Vector auxiliary and one scalar weighted constraint. On active blocks, $A \geq r(k + n - 1)/2$, use $h = n/2$, $c_0 = (k + 1)/2$ and local levels (the positive indices run from 1 to h , and the negative indices from 1 to n)

$$\begin{aligned} z_i^{+,L} &= r(c_0 + h - i), \\ z_i^- &= -r[c_0 + \min(i - 1, n - i)], \\ z_i^{+,R} &= r(c_0 + i - 1). \end{aligned}$$

Held white auxiliary inputs are $U_j = f(A - g_j z_j^{\text{loc}})$ and missing ones fA . On inactive blocks, prefix and tail set $U = 0$. Only uniformly finitely many early blocks are inactive. Local levels generate a dual rather than a claimed global primal path.

The baseline scalar direction has $\lambda_j^0 = (g_j f)^\top U_j$ on holds and zero on missing inputs. It has zero mass and support

$$h_\ell^0 = F \left\{ \frac{Ar}{2} n(n + 2k) - 4r^2 \sum_{j=0}^{n/2-1} (c_0 + j)^2 \right\}. \quad (21)$$

For $q_t(\lambda^0)$, the partial sums have the correct left-positive/right-negative signs, are zero at the negative center and at the end, and the local levels saturate $\Delta z = -r\Delta t \text{sign}(q_t(\lambda^0))$ whenever $q_t(\lambda^0) \neq 0$. Missing zero weights preserve the physical gap span $k + 1$. The held absolute levels have four copies of $r(c_0 + j)$, $j = 0, \dots, n/2 - 1$, so $\sum_{\text{held}} g_j z_j^{\text{loc}} = rn(n + 2k)/2$ and $\sum_{\text{held}} (z_j^{\text{loc}})^2 = 4r^2 \sum_{j=0}^{n/2-1} (c_0 + j)^2$. Their saturated support is $\sum \lambda_j^0 z_j^{\text{loc}}$ by summation by parts, giving (21). Here Δt is the full input separation between displayed held nodes; missing-slot interpolation keeps the same saturated slope along the partial-mass plateau.

Write $\gamma_\ell = (g_j f)_j$, $p_\ell = P_\ell \gamma_\ell$, and

$$\begin{aligned} N_\ell &= \gamma_\ell^\top P_\ell U_\ell, \quad D_\ell = \|p_\ell\|^2, \\ \alpha_\ell &= N_\ell/D_\ell, \quad v_\ell = P_\ell U_\ell - \alpha_\ell p_\ell. \end{aligned}$$

Apart from two gap-right inputs, the other $2n - 2$ held inputs are singletons with projection I_p and $\|\gamma_j\|^2 = F$. Thus $D_\ell \geq F(2n - 2) \geq Fn$. The difference $(\gamma_j^\top (P_\ell U)_j)_j - \lambda^0$ lies in two fixed spans only. In each span, $\sum_j |\gamma_j^\top (PU)_j| \leq \sqrt{mF} \|PU\| \leq mFA$; the baseline right-held point contributes at most FA . With $C_k = 2(k + 2)$ this gives

$$\begin{aligned} |N_\ell| &\leq C_k FA, \quad |\alpha_\ell| \leq C_k A/n, \\ \gamma_\ell^\top v_\ell &= 0, \quad P_\ell v_\ell = v_\ell, \\ \|v_\ell - P_\ell U_\ell\|^2 &= N_\ell^2/D_\ell \leq C_k^2 FA^2/n. \end{aligned}$$

The repaired mass is one scalar constraint, exactly matching one unrestricted scalar healthy level. It is not a coordinatewise vector-nuisance constraint.

For $\delta\lambda_j = \gamma_j^\top v_j - \lambda_j^0$, $\sum \delta\lambda_j = 0$ and $\sum |\gamma_j^\top p_j| \leq \|\gamma_\ell\| \|p_\ell\| \leq L_\ell F$. Hence $\|\delta\lambda_\ell\|_1 \leq C_k(k + 3)FA$. The full scalar rate class has support $r \sum_t |q_t(\lambda)|$ for a zero-mass vector; each partial sum is at most half its L_1 norm. A length- L_ℓ block correction therefore costs at most $r(L_\ell - 1)\|\delta\lambda_\ell\|_1/2 = O(FrA(n + k))$. Summation and restriction of the full path class give

$$\begin{aligned} h_{\mathcal{Z}_H}(\lambda) &\leq \sum h_\ell^0 + O(H^2) \\ &\leq \frac{Fr}{2} \sum A_\ell n_\ell^2 + O(H^2). \end{aligned} \quad (22)$$

The assembled v lies in range P and $\sum_j (g_j f)^\top v_j = 0$ exactly.

Observable dual and energy costs. The range condition makes v observable by the predictor pullback in Appendix A. Equation (17) bounds its deficit. On active inputs, auxiliary scalar errors are at most $(\eta + r)(n_\ell + k)$. The at most four tail errors are $O(H)$, prefix errors zero, and inactive blocks contribute

$O(1)$. Therefore $\|\theta - U\|^2 = O(H^2)$ and, by contraction and repair,

$$\|P\theta - v\|^2 \leq 2\|\theta - U\|^2 + 2C_k^2 F \sum A_\ell^2 / n_\ell = O(H^2).$$

The two input-gap centers lie at offsets $1/2 \pm (n_\ell + k)/2$. With fixed $\beta = \beta_k$ and fixed L_1, L_2 from (14), their exact oracle norm loss is

$$2\beta(A_\ell + \eta/2)^2 + \frac{\beta\eta^2}{2}(n_\ell + k)^2 + 4\eta(A_\ell + \eta/2)L_1 + 2\eta^2 L_2.$$

Consequently $\|(I - P)\theta\|^2 = 2\beta \sum A_\ell^2 + O(H^{3/2})$. The non-reflection cross has been retained, not asserted zero.

Substituting the support and energy bounds into (17) gives $2D \leq 2\beta \sum A_\ell^2 + Fr \sum A_\ell n_\ell^2 + O(H^2)$. Power summation yields $\sum A_\ell^2 = \eta^2 H^{5/2} / (5\sqrt{\xi}) + O(H^2)$ and $\sum A_\ell n_\ell^2 = \eta\sqrt{\xi} H^{5/2} / 5 + O(H^2)$. This proves (8). Minimizing at $\xi_* = 2\beta\eta / (Fr)$ matches (19), proving (7). $\square$

APPENDIX C

SCORE-PRESERVING QUADRATIC DEFICIT

Zero branch of Theorem 1.

Terminal dominance. An unread terminal move can be replaced by continued holding as in Appendix B; it suffices to consider calendars that read H . Their spans cover every post input.

Zero penalty: a profile-aware upper bound. Assume $\beta_k = 0$ and fix $L > n_0 + k + 2$. Keep the full-input cumulative task coefficient $M_t = \sum_{j=-n_0+1}^t g_j$. Hold the current sign s until $sM_t \geq L$. If at least $k + 2$ slots remain, execute the known exact- k move, flip s , and continue; otherwise hold to H . This is a deterministic calendar using the known profile at each planned start. It balances all input coefficients, including the missing ones, rather than just held signs.

The threshold overshoot is less than one and a move changes M_t by at most k . Thus the next opposite hold run reaches its threshold after between $2L - k$ and $2L + k + 2$ slots. The initial run exceeds two slots; a skipped final move adds at most $k + 1$ holds. For large H , every post hold run has at least two slots and

$$|M_t| \leq L + k + 2. \quad (23)$$

If N_h, N_m count post held inputs and moves, then $H = N_h + kN_m$ and $N_h \geq 2N_m$. The singleton count, excluding the right retained point of each gap, is $S = N_h - N_m \geq H/(k + 2)$.

Put $m = k + 1$. Since $P_m d_m = d_m$, only the affine slope is lost in each fixed-length gap:

$$E = \|(I - P)\theta\|^2 \leq N_m \eta^2 F m(m^2 - 1)/12 = O(H). \quad (24)$$

Let $\gamma = (g_j f)_j$, $w = P\theta$, $p = P\gamma$ and $\lambda_j = \gamma_j^\top w_j$. We have $\lambda_j = Fg_j a_j + e_j$, where $e_j = 0$ on singletons and is uniformly bounded on the fixed-length gaps independently of the amplitude. The post cumulative $S_t = \sum_{j=1}^t g_j = M_t - n_0$ has $|S_t| \leq L + k + 2 + n_0$ by (23). Abel's identity gives

$$\sum_{j=1}^t g_j a_j = a_t S_t - \eta \sum_{j=1}^{t-1} S_j.$$

All partial λ masses are therefore $O(t + 1)$ and $N = \sum_j \lambda_j = O(H)$. The fixed prefix changes only the constants.

Now $D = \|p\|^2$ lies between $FS \geq FH/(k + 2)$ and $F(H + n_0)$. Set $\alpha = N/D = O(1)$ and $v = w - \alpha p$. Then $Pv = v$, $\sum_j \gamma_j^\top v_j = 0$, and

$$\|\theta - v\|^2 = E + \alpha^2 D = O(H). \quad (25)$$

Each $\gamma_j^\top p_j$ is bounded by $F\sqrt{m}$ because all nonsingleton spans have fixed length. The repaired scalar weights $\lambda'_j = \gamma_j^\top v_j$ hence still have partial masses $O(t + 1)$. Equation (16) gives the complete-path support bound

$$h_{\mathcal{Z}_H}(\lambda') = r \sum_{t=-n_0+1}^{H-1} \left| \sum_{j=-n_0+1}^t \lambda'_j \right| = O(H^2). \quad (26)$$

The free level is handled by the exact repaired zero mass, not by fixing an initial healthy value. The observable range vector v satisfies (17), giving $D_{K,H}(\pi) \leq \|\theta - v\|^2/2 + h_{\mathcal{Z}_H}(\lambda')$. Equation (25), the support bound and (17) prove the $O(H^2)$ upper bound.

Zero penalty: an all-calendar lower bound. By monotonicity and (13), define finite

$$L_z = \max\{m \geq 1 : \beta_{m-1} = 0\} \geq k + 1, \\ b_+ = \inf_{m > L_z} \frac{\beta_{m-1}}{m} > 0.$$

The capacity bound gives $L_z \leq \nu_R$. Set $\epsilon = (4L_z + 2)^{-1}$ and $J_0 = \lceil \epsilon H \rceil$. Consider any calendar reading H .

If a positive-penalty interval ends at $t \geq J_0$, it lies entirely in the post affine region and has center amplitude $A \geq \eta(t + 1)/2 \geq \eta\epsilon H/2$, even if its span is long. For sufficiently large H , uniformly in the span, (14) gives $\|(I - P_m)\theta_m\|^2 \geq (b_+/2)mA^2$. Using $z = 0$ in the profile therefore gives

$$O_H - G_{K,H}(\pi) \geq \frac{1}{2} \|(I - P)\theta\|^2 \geq \frac{b_+ \eta^2 \epsilon^2}{16} H^2. \quad (27)$$

Otherwise every interval ending at or after J_0 has length at most L_z and preserves d_m . At least $(1 - \epsilon)H/L_z - O(1)$ such late intervals cover inputs $J_0, \dots, H$. Their center amplitudes are at least $\eta\epsilon H/2$ for large H . Choose the single complete healthy difference

$$z_j = (r/2)g_j, \\ \tilde{h}_j = fg_j z_j = (r/2)fg_j^2.$$

Because $|g_j - g_{j-1}| \leq 2$, this path obeys $|\Delta z| \leq r$, and the two paths $z/2, -z/2$ meet their individual rate bounds. The free prefix level makes the same choice legal there. For a late zero-penalty span, preservation of d_m gives

$$\langle P_m \theta_m, P_m \tilde{h}_m \rangle \\ = \frac{r}{2} F A \sum_{j=1}^m g_j^2 + \eta \langle P_m d_q, P_m \tilde{h}_m \rangle \\ \geq \frac{r}{2} F A - C_0.$$

Here C_0 is fixed since $m \leq L_z$. The right retained input has $g^2 = 1$, so $\sum g_j^2 \geq 1$. No entrywise positivity of the projection is assumed. The total late cross is thus at least

$$\frac{rF\eta\epsilon(1-\epsilon)}{4L_z}H^2 - O(H).$$

All early intervals end before J_0 . Contraction and Cauchy bound their cross from below by $-(r/2)F \sum_{\text{early}} \sqrt{m \sum a_j^2}$. Their total input count is at most $J_0 - 1 + n_0$, and $a_j \leq \rho_0 + \eta J_0$, so the early contribution is at least $-(r/2)F\eta\epsilon^2H^2 - O(H)$. Finally $\|P\tilde{h}\|^2 \leq r^2F(H+n_0)/4 = O(H)$. Evaluating the profiled infimum at this legal full path and expanding the half square gives

$$\begin{aligned} O_H - G_{K,H}(\pi) &\geq \frac{1}{2}\|(I-P)\theta\|^2 + \\ &\quad \langle P\theta, P\tilde{h} \rangle - \frac{1}{2}\|P\tilde{h}\|^2 \\ &\geq \frac{rF\eta}{2} \left[\frac{\epsilon(1-\epsilon)}{2L_z} - \epsilon^2 \right] H^2 - O(H) \\ &= \frac{rF\eta\epsilon}{8L_z} H^2 - O(H). \end{aligned} \quad (28)$$

The last equality uses $\epsilon = (4L_z + 2)^{-1}$. All constants in both cases are independent of the calendar. A smaller positive constant than the minimum of the coefficients in (27) and (28) gives the lower bound in (9). Together with the profile-aware upper construction, this completes the two-regime theorem. $\square$

APPENDIX D

CONTROLLER DESIGN PROOFS

A. Proof of Proposition 2

Proof. For fixed $m = k+1$ the explained constant-mode norm is $R_m = (1+c)(1-c^m)/[(1-c)(1+c^m)]$. Its logarithmic derivative is positive because $\sum_{j=0}^{m-1} c^{2j} > mc^{m-1}$ by strict AM-GM. The exact increment in k is

$$\kappa_c(k+1) - \kappa_c(k) = \frac{(B_m - A_m c^m)^2}{B_m(B_m + c^{2m})} > 0.$$

The plateau $s \geq 2$ is $(\gamma^{1/(s-1)}, \gamma^{1/s}]$; the first plateau has right endpoint γ . Each nonempty intersection with the compact controller interval contains its rightmost point. Fixed- k monotonicity places its minimum there. At the equality $c^s = \gamma$ the shorter s applies, so a floating-point logarithm must not change the tie rule. Finally $m\delta = L + O(\delta)$ and $R_m = \coth(\delta/2) \tanh(m\delta/2)$. The limit follows, and positivity is $L - 2 \tanh(L/2) = \int_0^L \tanh^2(t/2) dt > 0$. $\square$

B. Proof of Theorem 2

Proof. Continuity of the spectral radius on the compact family gives $a_* := \max_K \rho(A_K) < 1$. Choose $a_* < \tau < 1$. The resolvent has a finite norm maximum R_* on $\mathcal{K} \times \{|z| = \tau\}$. Matrix Cauchy integration gives $\|A_K^j\| \leq R_* \tau^{j+1}$, hence a common finite bound $S_* \geq \sum_j \|A_K^j B\|$. Since $W_{m,K} \succeq Q$, the affine-cross constant in Appendix B is uniformly bounded by $C_* := \|Q^{-1}\| S_*^2$.

Each $\beta_K(\ell)$ is continuous and strictly positive for fixed $1 \leq \ell \leq k_{\max}$. Finite k and compactness give common $0 < \beta_{\text{lo}} \leq \beta_K(k(K)) \leq \beta_{\text{hi}}$. The tail $\beta_K(\ell) \geq (\ell+1)F - C_*$ and monotonicity give a common positive b_* below all $\beta_K(\ell)/\ell$, $\ell \geq 1$. For fixed ε, δ , the converse proof therefore uses one absorption threshold and one $o(H^{5/2})$ remainder for all gains and calendars. Its lower coefficient is $\sqrt{1-\delta}(1-\varepsilon^{5/2})C(K)$. First choose ε, δ using $\sup C(K) < \infty$, then the common horizon threshold; this gives the uniform lower comparison.

The optimal $\xi_K = 2\beta_K(k(K))\eta/(Fr)$ lies in a fixed positive compact interval. Packing has a common bounded padding, because the unused time is less than the next block length and is evenly distributed. Thus $n_l = \xi_K l + O(1)$ and $A_l = \eta \xi_K l^2 + O(l+1)$ uniformly. Because the base total duration is $\xi M(M+1) + O(M)$ uniformly and the unused budget is less than the next block duration, evenly spreading its four-slot units adds one common bounded even amount to each n_l . A common finite early cutoff makes every later block active: the center amplitude is at least $\eta \xi_{\text{lo}} l(l-1)$, whereas the active threshold is at most $r(k_{\max} + \xi_{\text{hi}} l + C_n)/2$ for one common padding constant C_n . The quadratic eventually dominates the linear bound, giving a common finite inactive prefix. Auxiliary error, repair energy and support correction are all bounded by common multiples of $\sum(n_l+k)^3$, $\sum A_l^2/n_l$ and $\sum A_l(n_l+k)$, each $O(H^2)$. The fixed spans have common affine coefficients, and the two power sums in Appendix B have common $O(H^2)$ errors. Hence $D_H^{*,K} \leq C(K)H^{5/2} + C_U H^2$ for one C_U . This proves uniform convergence. Taking infima changes any uniformly bounded error by no more than its uniform bound, yielding the joint statement. $\square$

APPENDIX E

BOUNDED TASK-EVEN PERTURBATION PROOF

Proof. In the real input projection let $t = P(fa)$ and $\mathcal{K} = \{P(fgz) : z \in \mathcal{Z}_H\}$. Each extra mean $P(fe)$ has norm at most $R = E\sqrt{FN}$. For $d = \text{dist}(t, \mathcal{K})$, class inclusion and the triangle inequality give $(d-R)_+ \leq d_{\text{mix}} \leq d$. Since $0 \in \mathcal{K}$, $d \leq \sqrt{F}A_H$. For $G = d^2/2$, separately considering $d \geq R$ and $d < R$ gives $0 \leq G_{\text{odd}} - G_{\text{mix}} \leq dR \leq FE\sqrt{N}A_H$. Taking suprema over the same calendars proves the deficit comparison. $\square$

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